

SciResearcher: Scaling Deep Research Agents for Frontier Scientific Reasoning

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Abstract

Frontier scientific reasoning is rapidly emerging as a key foundation for advancing AI agents in automated scientific discovery. Deep research agents offer a promising approach to this challenge. These models develop robust problem-solving capabilities through post-training on information-seeking tasks, which are typically curated via knowledge graph construction or iterative web browsing. However, these strategies face inherent limitations in frontier science, where domain-specific knowledge is scattered across sparse and heterogeneous academic sources, and problem solving requires sophisticated computation and reasoning far beyond factual recall. To bridge this gap, we introduce *SciResearcher*, a fully automated agentic framework for frontier-science data construction. SciResearcher synthesizes diverse conceptual and computational tasks grounded in academic evidence, while eliciting information acquisition, tool-integrated reasoning, and long-horizon capabilities. Leveraging the curated data for supervised fine-tuning and agentic reinforcement learning, we develop *SciResearcher-8B*, an agent foundation model that achieves 19.46% on the HLE-Bio/Chem-Gold benchmark, establishing a new state of the art at its parameter scale and surpassing several larger proprietary agents. It further achieves 13–15% absolute gains on SuperGPQA-Hard-Biology and TRQA-Literature benchmarks. Overall, SciResearcher introduces a new paradigm for automated data construction for frontier scientific reasoning and offers a scalable path toward future scientific agents.

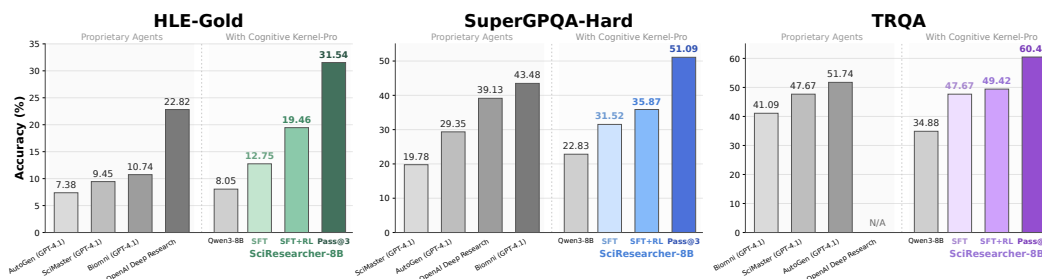


Figure 1: Performance comparison on HLE-Bio/Chem-Gold ($n = 149$), SuperGPQA-Hard-Biology ($n = 92$), and TRQA-Literature ($n = 172$). Our agent foundation model, *SciResearcher-8B*, outperforms existing proprietary scientific agents and is comparable to OpenAI Deep Research.

*Work done during Tianshi’s internship at Tencent.

1 Introduction

Frontier scientific reasoning captures an AI system’s ability to solve challenging, expert-level scientific problems at the boundaries of human knowledge [29, 38]. Such problems arise in scientific domains where relevant knowledge is often incomplete, rapidly evolving, and distributed across diverse sources [11, 5]. As AI-driven scientific discovery systems such as LLM-powered research assistants [30, 33] and autonomous AI scientists [21, 25, 50] become increasingly prevalent and capable, reasoning effectively and reliably in these applications is becoming critical.

Deep research agents [27, 4, 9] have emerged as a promising approach to frontier scientific reasoning, benefiting from their ability to acquire up-to-date knowledge through real-time web search and to carry out long-horizon tasks. A central strategy for advancing deep research agents is agent post-training on information-seeking tasks. For general factual knowledge, automated construction of information-seeking tasks has largely followed two paradigms: (1) knowledge graph construction-based methods [17, 35, 16], which first build a structured graph over entities or webpages and then sample paths or subgraphs to synthesize tasks; and (2) agent-based methods [44, 45, 19], which instead allow an agent to iteratively search and browse from seed entities or URLs to construct a local information space for task synthesis. Existing instantiations of both paradigms, however, are largely grounded in Wikipedia-like, entity-centric factual knowledge, and many of their automatically constructed tasks emphasize short-answer, attribute-, or fact-seeking supervision.

While these paradigms have achieved strong results for general-domain information seeking, they are inherently limited in their applicability to frontier scientific reasoning (Figure 2). First, frontier scientific entities are associated with heterogeneous, noisy, and highly context-dependent ontologies [49, 7]. Unlike general-domain entities, which are often described by relatively standardized attributes, scientific entities rarely admit a shared attribute schema, and even similar attributes require substantial domain-specific context to be meaningful. This challenges the feasibility of structured graph construction. Second, knowledge relevant to frontier scientific problems is often sparse and fragmented across loosely connected academic sources, rather than organized as densely interlinked webpages [2, 32]. As a result, frontier scientific concepts rarely have a canonical entry point analogous to a Wikipedia page, making continuous web traversal difficult to realize effectively. Moreover, frontier scientific reasoning often requires nontrivial computation over complex scenarios and scientific models [29, 38, 26], which is not captured by deep research task construction approaches.

To address these limitations, we introduce *SciResearcher*, a fully automated data construction framework for frontier scientific reasoning. The framework operates over heterogeneous scientific concepts, integrates evidence scattered across weakly connected sources, and treats scientific computation as a core component of reasoning. Built on a curated pool of frontier scientific entities, it comprises two data construction pipelines, targeting *conceptual* and *computational* questions, respectively. For conceptual task construction, we employ web agents to browse from seed scientific entities, gather academic evidence, and generate rich, grounded questions. We then apply an iterative anchor²-based task augmentation stage to increase task complexity. For computational task construction, we perform a three-level evidence selection process to identify novel and challenging computational models (typically comprising governing equations, mechanistic understanding, and required input parameters) associated with seed entities, and then generate questions and obtain reference answers through voting-based solver verification. Leveraging this framework, we construct SciResearcherQA, a

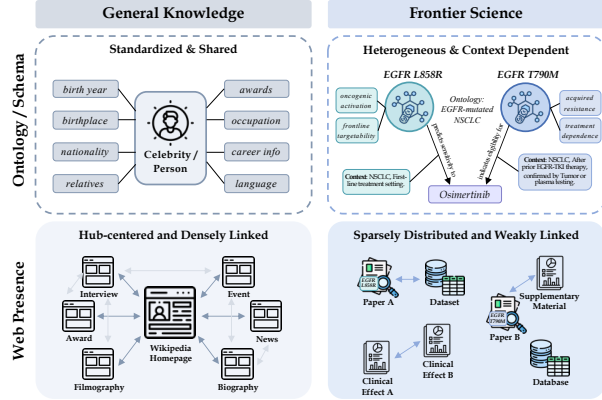


Figure 2: Comparison of ontology and web presence between general knowledge and frontier science.

²We define an *anchor* as the key, decisive scientific entity that plays a central role in a question, serving as its signature referent and the primary handle for reasoning about the problem.

challenging scientific reasoning dataset that stress-tests long-horizon, multi-evidence problem solving grounded in frontier knowledge and computation.

Using SciResearcherQA, we perform agent post-training following an established recipe that begins with supervised fine-tuning using rejection sampling and followed by reinforcement learning via GRPO [31]. Based on Qwen3-8B [47], we train *SciResearcher-8B*, which achieves strong performance across multiple frontier scientific reasoning benchmarks, including HLE-Bio/Chem-Gold [43], SuperGPQA-Hard-Biology [23], and TRQA-Literature [48]. On HLE-Bio/Chem-Gold, our agent attains 19.46% pass@1 and 31.54% pass@3, surpassing existing proprietary agents such as SciMaster [3] with GPT-4.1 backbone and approaching the performance of frontier proprietary deep research systems such as OpenAI Deep Research [27]. Moreover, SciResearcher-8B yields substantial gains on SuperGPQA-Bio-Hard and TRQA-Literature, improving absolute performance by 13.04% and 14.54%, respectively.

Further analysis shows that post-training leads to substantially more extended and tool-intensive reasoning behavior: compared with the baseline, our agent produces trajectories that are $0.3\text{--}2.7\times$ longer and exhibits correspondingly higher tool-use frequency. Interestingly, reinforcement learning induces an adaptive effect relative to the SFT checkpoint: on the more challenging HLE-Bio/Chem-Gold benchmark, the agent learns to allocate more steps and tool calls, whereas on relatively easier benchmarks it uses slightly fewer steps while still achieving stronger performance. These results suggest that SciResearcherQA improves frontier scientific reasoning performance together with cultivating more adaptive long-horizon information-seeking behavior.

Overall, SciResearcher offers a new perspective on automated data construction, extending it from general factual information-seeking tasks to frontier scientific reasoning. Our results highlight the promise of automatically constructed scientific training data for scaling deep research agents toward more capable and reliable scientific problem solving.

2 The *SciResearcher* Data Construction Framework

In this section, we introduce *SciResearcher*, a fully automated data construction framework (Figure 3) for frontier scientific reasoning tasks. Given a curated set of seed entities, SciResearcher generates both conceptual and computational reasoning tasks grounded in academic evidence through agentic web exploration. Examples of the generated questions are shown in Table 1.

2.1 Seed Entity Acquisition

High-quality, domain-specific scientific entities are essential for constructing frontier scientific tasks. To obtain them, we implement a three-stage seed entity acquisition pipeline. First, we extract an initial pool of scientific entities from public datasets, together with their annotated domain and ontology information. Second, we use LLMs to generate additional entities within the same ontology, thereby expanding the candidate pool. Third, we automatically score all entities in the augmented pool using three metrics—frontier relevance, concreteness, and specificity—to filter for quality. Finally, we retain the top 5% of entities as seed entities for downstream task curation.

2.2 Conceptual Task Curation

The detailed curation pipeline for conceptual tasks is illustrated in Figure 3(a). Starting from a target seed entity, we employ a web agent with a proprietary LLM backbone to iteratively search for and browse academic sources on the web, with the goal of constructing a base conceptual question grounded in verifiable scientific evidence. Concretely, the agent first performs iterative scout searches to identify promising academic sources relevant to the seed entity. It then uses a `url2evidence` sub-agent to access the selected paper, extract the key supporting evidence, and formulate an initial conceptual multiple-choice question together with plausible confounders. This initial question serves as the semantic backbone for subsequent augmentation.

To increase task complexity beyond single-hop retrieval, we further perform anchor-based question augmentation. Given the current question, we first extract candidate anchor entities from the question text and evaluate them using three criteria: whether the entity is decisive for deriving the final answer, whether it is decoupled from the surface form of the answer options, and whether it is sufficiently

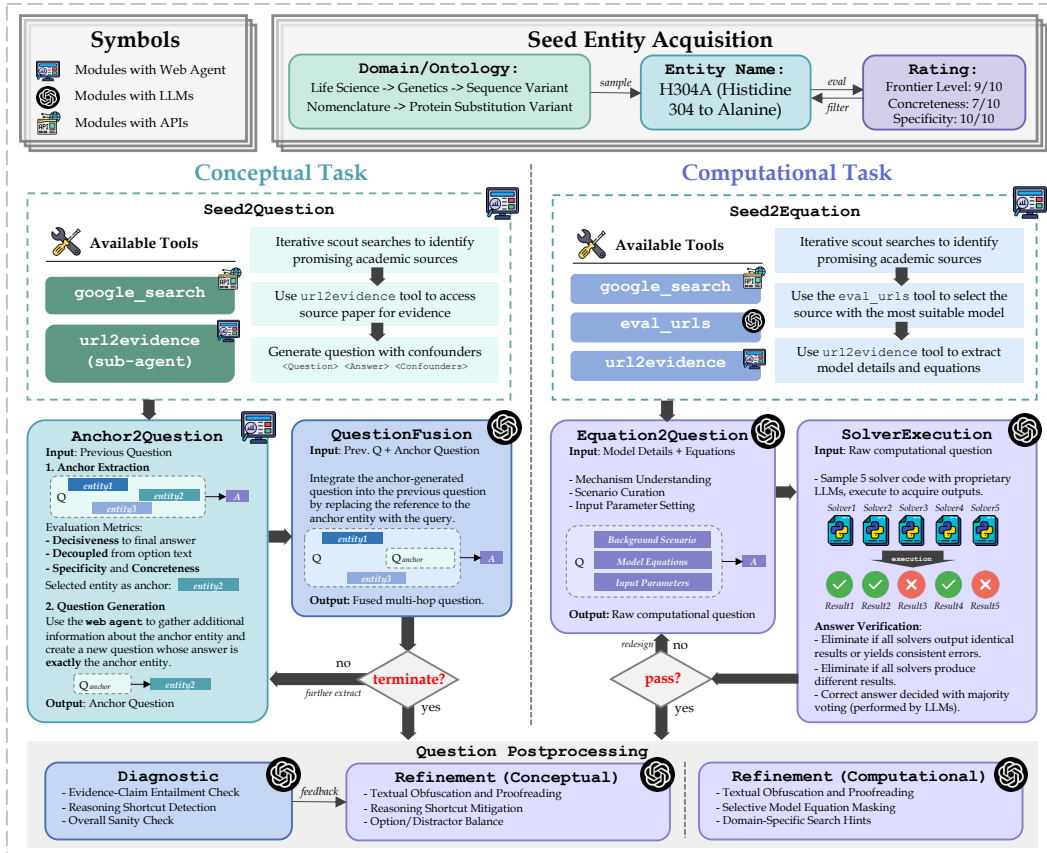


Figure 3: Overview of our *SciResearcher* data construction framework.

specific and concrete to support further evidence-grounded expansion. After selecting the best anchor, we invoke a new web agent instance to gather additional academic evidence about that anchor and generate a new question whose answer is exactly the anchor entity. This newly generated question is then fused back into the previous question by replacing the original anchor mention, thereby converting a direct clue into an additional reasoning step. The augmentation process can be repeated multiple times, recursively transforming a seed question into a multi-hop question that requires long-horizon browsing and evidence aggregation across multiple independent sources. Figure 4 shows a running example of this question evolution process.

Unlike prior agent-based approaches such as WebDancer [44] and WebExplorer [19], our conceptual task curation instantiates each augmentation step with a separate web agent instance. This design reduces dependence on a single search trajectory and more explicitly stress-tests an agent’s ability to perform multi-source academic retrieval, cross-source integration, and compositional reasoning.

2.3 Computational Task Curation

The curation pipeline for computational tasks is illustrated in Figure 3(b). In contrast to conceptual tasks, which primarily evaluate information seeking and synthesis, computational tasks additionally require agents to apply retrieved scientific knowledge to perform nontrivial quantitative reasoning.

The pipeline begins with a tailored web agent that identifies and extracts the most appropriate advanced computational model associated with the seed entity through a three-level evidence selection process. First, multiple scouting searches are performed to identify promising sources based on their titles and content snippets. Second, the selected links are evaluated and filtered using the `eval_urls` tool, which applies four metrics—model exclusiveness, search identifiability, computational complexity, and LLM unfamiliarity—to support comprehensive assessment. Third, sub-agents are deployed to conduct a deep dive into the final selected URLs, extracting the complete model specification together with the scenarios and constraints required for its application.

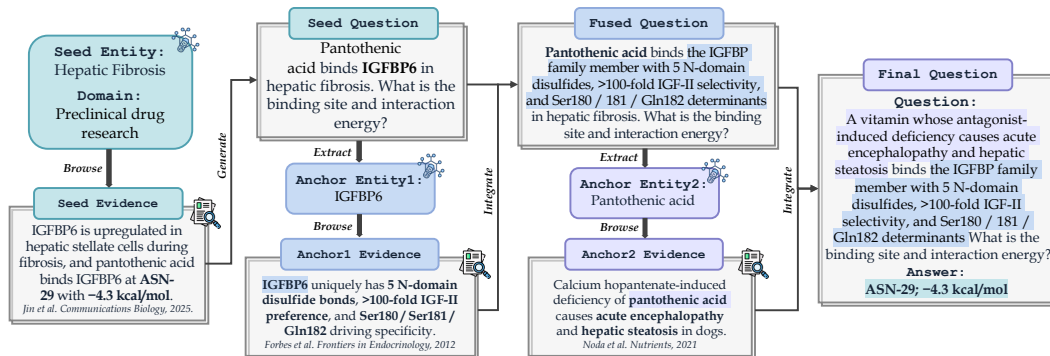


Figure 4: A running example of a question evolution pipeline for conceptual task curation. Question fusion and postprocessing details are omitted.

Based on the extracted model, the system constructs a scenario-based computational question. This process requires understanding the scientific mechanism encoded by the model, curating a realistic background scenario, and specifying the necessary input parameters. The resulting question therefore tests whether an agent can not only retrieve the relevant scientific source, but also instantiate the model correctly in a concrete setting.

Because such generated questions do not come with a guaranteed correct answer, we further perform answer acquisition and verification. Specifically, we sample five candidate Python solvers from proprietary LLMs and execute them to obtain candidate outputs. We then filter out low-quality questions based on solver agreement patterns: questions are rejected if all solvers return the same result, if all solvers yield consistent errors, or if all solvers produce different answers. These cases typically indicate that the question is respectively too easy, not executable, or too unstable. For the remaining questions, the final answer is determined by majority voting over solver outputs, followed by LLM-based verification. Questions that fail this verification process are sent back for redesign.

2.4 Question Postprocessing

After generating raw questions, we apply a postprocessing stage to improve question quality, reduce shortcut-based answering, and increase the amount of retrieval and reasoning required to solve each task. We begin with a general diagnostic pass, which includes evidence–claim entailment checking, reasoning shortcut detection, and overall sanity checking. Based on diagnostic feedback, we then apply category-specific refinement. For conceptual questions, we perform textual obfuscation and proofreading, mitigate reasoning shortcuts, and improve the balance of answer options and distractors. For computational questions, we additionally perform selective masking of model equations and inject domain-specific search hints to encourage agents to retrieve the relevant scientific source and reconstruct the appropriate computational model.

2.5 Analysis on SciResearcherQA

Figure 5(a) visualizes the most frequent words in the two question types of SciResearcherQA. Both word clouds contain domain-specific terminology from biology and chemistry, such as *cell*, *reaction*, and *drug*. Conceptual questions tend to emphasize more qualitative terms, such as *effect*, *dependent*, and *deficiency*, whereas computational questions feature more quantitative terms, such as *rate*, *concentration*, and *parameter*.

Figure 5(b) further presents the trajectory distribution of an agent using Claude-Sonnet-4.5 as the backbone. On conceptual questions, the agent achieved an accuracy of 74.9% with an average of 7.74 macro steps³, whereas on computational questions it achieved 45.1% accuracy with an average of 9.14 macro steps. These results highlight both the substantial difficulty of the benchmark and its long-horizon reasoning requirements. Compared with conceptual questions, computational questions are markedly more challenging for current agents, requiring more reasoning steps on average while yielding substantially lower accuracy.

³Since we use the multi-agent framework of Cognitive Kernel-Pro, a macro step denotes one planning step and one action step of the main agent. On average, one macro step corresponds to 4.1–4.9 total LLM calls in our experiments.

Table 1: Examples of questions and supporting evidence for both question types in SciResearcherQA. Supporting evidence and its corresponding question content are highlighted using the same color.

Type	Example Question	Supporting Evidence
Conceptual	A vitamin whose deficiency causes concurrent acute encephalopathy and hepatic steatosis when induced by a specific antagonist binds to a protein with the following properties: (1) contains only 5 N-terminal domain disulfide bonds, unlike other family members with 6; (2) exhibits more than 100-fold selectivity for IGF-II over IGF-I; and (3) has C-terminal residues Ser180, Ser181, and Gln182 determining its IGF-II specificity. What is the binding site and interaction energy? Answer: ASN-29 with binding energy of -4.3 kcal/mol	<i>Source 1:</i> Vitamins and Minerals for Energy, Fatigue and Cognition: A Narrative Review of the Biochemical and Clinical Evidence (<i>Nutrients</i>, 2020). <i>Source 2:</i> Insulin-Like Growth Factor Binding Proteins: A Structural Perspective (<i>Front. Endocrinol.</i>, 2012). <i>Source 3:</i> Pantothenic acid ameliorates hepatic fibrosis by targeting IGFBP6 to regulate the TGF-β/SMADs pathway (<i>Commun. Biol.</i>, 2025).
Computational	A 42-year-old patient with a WHO Grade II, 1p/19q-codeleted oligodendroglioma is treated with temozolomide (TMZ) chemotherapy. Let $P(t)$ denote the volume of proliferative tumor cells, $D(t)$ the volume of lethally damaged cells, and $C(t)$ the intratumoral TMZ concentration, with total tumor volume $V(t) = P(t) + D(t)$. The system follows the standard compartmental treatment-response equations with logistic growth limitation and two TMZ-induced effects. (Initial Conditions Omitted) The patient completes three full treatment cycles (84 days total), with dosing on days 1, 2, 3, 4, 5, 29, 30, 31, 32, 33, 57, 58, 59, 60, 61. What is the total tumor volume $V(84)$ in cm^3, immediately after completion of the third cycle? Answer: 62.77 cm^3	<i>Source:</i> Computational design of improved standardized chemotherapy protocols for grade II oligodendrogliomas (<i>PLOS Comput. Biol.</i>, 2019). Model equations (ODE system): $\frac{dP}{dt} = \rho P \left(1 - \frac{P+D}{K} \right) - \alpha_1 PC - \alpha_2 PC,$ $\frac{dD}{dt} = \alpha_1 PC - \frac{\rho}{\kappa} D \left(1 - \frac{P+D}{K} \right),$ $\frac{dC}{dt} = -\lambda C.$

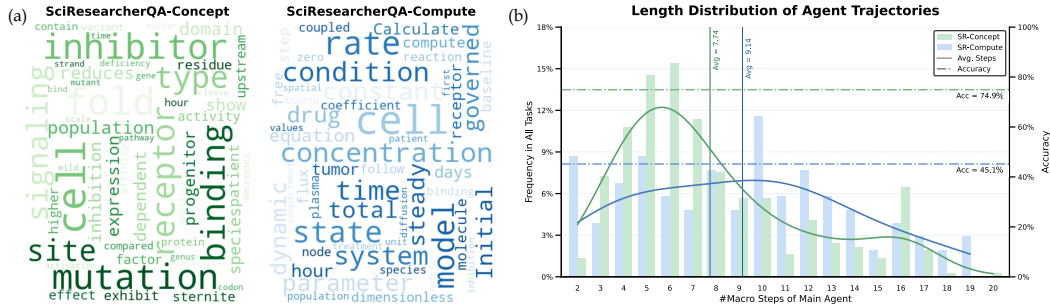


Figure 5: (a) Word clouds of the curated questions from the two pipelines. (b) Distribution and performance of Claude-Sonnet-4.5 across the two question types.

3 Experiments

3.1 Experimental Setups

Deep Research Agent Framework In our experiments, we adopt Cognitive Kernel-Pro [9] as the foundation of our deep research agent. CK-Pro is a multi-agent framework optimized for GAIA-like [24] information-seeking tasks, consisting of a main agent that coordinates two specialized sub-agents for web browsing and file analysis. We further adapt the framework at the prompt and instruction levels to better align it with frontier scientific domains, resulting in a 4–8% performance improvement in preliminary experiments over the original framework.

Benchmarks Following prior work [3, 34], we evaluate our method on three frontier scientific reasoning benchmarks: 1) **HLE-Bio/Chem-Gold** [43] is an expert-verified subset of Humanity’s Last Exam [29], comprising 149 highly challenging questions in advanced biology and chemistry. 2) **SuperGPQA-Hard-Biology** [23] contains 92 expert-annotated biology questions that emphasize difficult, reasoning-intensive scientific problem solving. 3) **TRQA-Literature** [48] is a knowledge-intensive benchmark grounded in advanced therapeutic research literature. Together, these benchmarks provide complementary coverage of frontier scientific reasoning, ranging from research-level scientific understanding to literature-grounded multi-step inference.

Table 3: Performance comparison on HLE-Bio/Chem-Gold (n=149), SuperGPQA-Hard-Biology (n=92), and TRQA-Literature (n=172). Results of proprietary agent baselines are reported by Tang et al. [34]. *Note: In TRQA experiments, we exclude TRQA from the training data of SciResearcher.

Agent Framework	LLM Backbone	HLE-Gold	SuperGPQA-Hard	TRQA*
<i>Vanilla LLMs</i>				
–	Qwen3-32B	5.37	31.52	37.79
–	Kimi-K2	6.71	48.91	38.37
–	Deepseek V3.1	13.42	66.30	43.60
–	Gemini-2.5 Pro	18.79	65.22	45.93
<i>Agent Systems</i>				
AutoGen [46]	GPT-4.1	7.38	29.35	51.74
SciMaster [3]	GPT-4.1	9.45	19.78	47.67
Biomni [13]	GPT-4.1	10.74	43.48	41.09
OpenAI Deep Research [27]	o4-mini	22.82	39.13	-
Cognitive Kernel-Pro [9]	Qwen3-8B	8.05	22.83	34.88
	Qwen3-32B	10.74	38.04	46.51
	<i>SciResearcher</i> -8B-SFT	12.75	31.52	47.67
	<i>SciResearcher</i> -8B-RL	19.46	35.87	49.42
	-pass@3	31.54	51.09	60.47

Data and Training The composition of the training data is summarized in Table 2. In addition to the two data types generated by SciResearcher, we introduce two auxiliary data sources to improve distributional balance. First, we incorporate a small subset of SciBench [40] as a source of relatively simple scientific reasoning questions, which helps offset the difficulty of SciResearcherQA-Compute and reduce overthinking during training. Second, we include TRQA as a source of multiple-selection MCQs to counterbalance the predominance of single-selection MCQs in SciResearcherQA-Concept. For evaluations on TRQA-Literature, we train a separate checkpoint with TRQA removed from the training mixture.

We train our agent foundation model, *SciResearcher*-8B, based on Qwen3-8B [47], following the standard two-stage training paradigm of cold-start supervised fine-tuning (SFT) followed by reinforcement learning. In the first stage, we collect agent trajectories using Claude-Sonnet-4.5 [1] as the teacher model and perform supervised fine-tuning with rejection sampling to initialize the model’s tool-use and long-horizon decision-making abilities. In the second stage, we further optimize the model with reinforcement learning using the GRPO algorithm [31] with outcome-only rewards, encouraging the agent to discover more effective task-completion strategies through interaction.

Our training objective is to improve the main agent’s ability to carry out long-horizon scientific tasks, particularly in planning, tool use, and multi-step execution. Accordingly, we train only on trajectories produced by the main agent. The web-browsing and file-analysis sub-agents are kept frozen throughout training and are treated as external tools rather than trainable components.

3.2 Results and Analyses

Main Result The main experimental results are reported in Table 3. Overall, *SciResearcher*-8B yields substantial improvements over the base Qwen3-8B agent across all three frontier scientific reasoning benchmarks, demonstrating the effectiveness of our overall approach. On HLE-Bio/Chem-

Table 2: Training data composition. # **Tasks** indicates number of QA pairs and # **Steps** indicates number of step-level messages for training.

Dataset	# Tasks	# Steps
SciResearcherQA-Concept	371	2,872
SciResearcherQA-Compute	104	951
TRQA-Literature [48]	172	932
SciBench [40]	80	350
Total	727	5,105

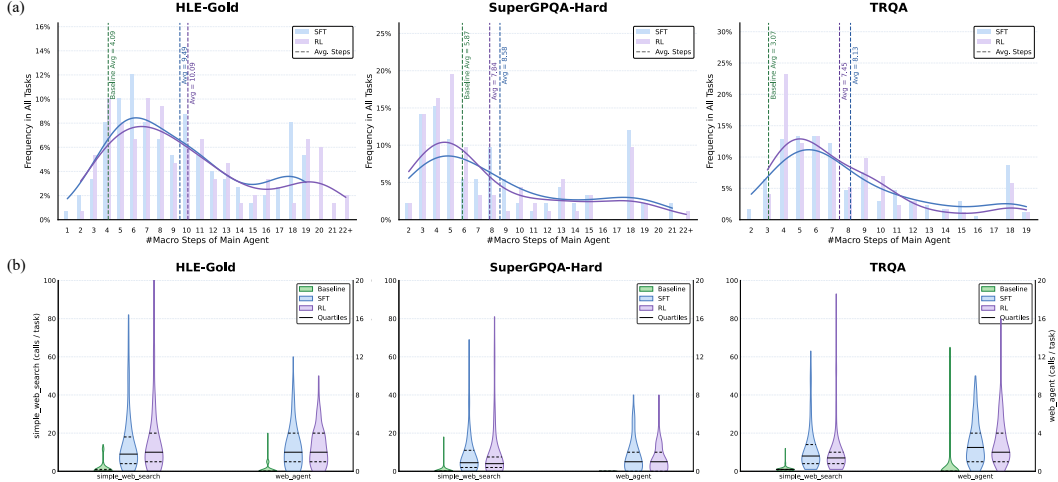


Figure 6: (a) Distribution of trajectory lengths (in macro steps) for SFT and RL checkpoints. (b) Distribution of tool-use frequency for simple web search and web agent.

Table 4: Ablation study of training data configurations (cumulative) on two benchmarks.

Training Setting	HLE-Gold	SuperGPQA-Hard
Qwen3-8B (baseline)	8.05	22.83
+ SciResearcherQA-Concept	10.74 (+2.69)	25.00 (+2.17)
+ SciResearcherQA-Compute	12.08 (+1.34)	28.26 (+3.26)
+ TRQA & SciBench	12.75 (+0.67)	31.52 (+3.26)

Gold, *SciResearcher*-8B-RL achieves 19.46%, improving over the Cognitive Kernel-Pro baseline by more than 11 absolute points and clearly outperforming prior proprietary scientific agent systems such as SciMaster and Biomni, both built on GPT-4.1. Although it remains below OpenAI Deep Research with o4-mini, our result already approaches that of frontier proprietary deep research systems despite using a much smaller open 8B backbone. On SuperGPQA-Hard-Biology and TRQA-Literature, *SciResearcher*-8B-RL reaches 35.87% and 49.42%, corresponding to absolute gains of 13.04 and 14.54 points over the baseline, respectively. These results indicate that our training setup consistently improves performance on both literature-grounded conceptual reasoning and more demanding scientific problem solving. We also observe significant gains under *pass@3* evaluation, suggesting additional headroom for performance improvement through inference-time scaling.

Ablation Study We further conduct an ablation study to evaluate the contribution of each training data component (in the supervised fine-tuning stage). As shown in Table 4, performance improves consistently on both HLE-Gold and SuperGPQA-Hard as additional data sources are introduced during supervised fine-tuning, indicating that each component provides complementary supervision. In particular, SciResearcherQA-Concept contributes gains of 2.17–2.69 points across the two benchmarks, while SciResearcherQA-Compute yields a further 1.34–3.26 points, demonstrating the value of both conceptual and computational task curation. Incorporating TRQA and SciBench leads to additional improvements, further validating the design of the overall training mixture. Taken together, these results show that the two proposed SciResearcherQA components serve as the core source of performance gains, while the auxiliary data provides useful complementary coverage.

Long-Horizon and Tool-Use Figure 6(a) shows the distribution of trajectory lengths for the baseline, SFT, and RL agents across the three benchmarks. After training, the agent generally produces substantially longer trajectories than the baseline, with trajectory lengths increasing by roughly $0.3\times$ to $2.7\times$, indicating a markedly stronger tendency to sustain multi-step exploration and reasoning. A notable pattern emerges when comparing SFT and RL: on the most difficult benchmark, HLE-Gold, the RL checkpoint tends to generate even longer trajectories than the SFT checkpoint, whereas on the other two comparatively easier benchmarks it produces shorter trajectories. This

suggests that reinforcement learning does not simply encourage more steps indiscriminately, but may instead improve the agent’s ability to allocate search and reasoning effort more adaptively based on task difficulty.

Figure 6(b) provides a complementary view through tool-use statistics, comparing the usage distributions of simple web search and the web-agent tool. Across all three benchmarks, both the SFT and RL checkpoints invoke tools substantially more often than the baseline, indicating that training improves the agent’s willingness and ability to rely on external information sources rather than prematurely answering from parametric memory alone. Furthermore, the RL checkpoint typically attains higher maximum tool-use counts than the SFT checkpoint, suggesting a stronger capacity to maintain extended tool-assisted reasoning chains when necessary. Taken together, these findings show that the gains of *SciResearcher*-8B are accompanied by measurable changes in behavior: the model not only answers more accurately, but also exhibits more persistent long-horizon planning and more intensive tool use.

4 Related Work

4.1 Deep Research Agents and Data Construction

Initial progress in deep research agents has been led by proprietary systems built on frontier models [27, 4, 28]. In the open-source literature, the main thrust has been automated data curation for agentic post-training, rather than agent orchestration alone. Existing approaches broadly fall into retrieval-centric pipelines, which synthesize supervision from graph construction, crawling, or traversal over webpages and entities [17, 16, 45, 36], and exploration-centric pipelines, which let agents iteratively search, browse, and expand local information spaces to construct harder long-horizon tasks and trajectories [35, 44, 19, 39]. While these efforts have substantially improved open agents’ ability to perform multi-step web information seeking, most synthesized supervision remains centered on general-domain, web-grounded retrieval, often with short-answer or fact-seeking targets. A parallel line of work improves deep research agents through self-evolving or self-improving mechanisms [8, 18, 37, 12]. Finally, several recent benchmarks have been proposed to stress-test long-horizon information seeking and research-style agent behavior [41, 10, 6].

4.2 Agents in Scientific Reasoning and Discovery

Many previous works improve the scientific reasoning of LLMs by designing specialized agent frameworks and integrating domain-specific knowledge sources. SciMaster [3] studies general scientific reasoning with parallel solution sampling and iterative refinement. Biomni [13] supports biomedical reasoning and discovery with tailored tool use and domain-specific databases. Eigen-1 [34] targets biology and chemistry with adaptive retrieval-augmented generation over a curated scientific paper corpus. Meanwhile, as LLM-based agents become more capable, researchers have increasingly applied them to scientific research and discovery [50, 22, 51]. These efforts range from iterative optimization agents that automatically improve the performance of a specific machine learning model based on execution feedback [14, 20, 15] to fully autonomous AI scientists that carry out the research loop from ideation to paper writing [21, 42].

5 Conclusion

We presented *SciResearcher*, a fully automated data construction framework for frontier scientific reasoning that moves beyond general-domain information-seeking supervision toward tasks grounded in heterogeneous academic evidence, scientific computation, and long-horizon reasoning. Through tailored curation pipelines for both conceptual and computational questions, *SciResearcher* generates high-quality scientific reasoning tasks that support agent post-training. The resulting model, *SciResearcher*-8B, achieves remarkable performance across multiple frontier scientific reasoning benchmarks, outperforming several proprietary scientific agents and exhibiting more adaptive, tool-intensive reasoning behavior. More broadly, our work offers a new perspective on fully automated data construction for frontier scientific reasoning, highlighting the promise of tailored question typologies as a scalable training paradigm for scientific agents. Future work may focus on investigating the fundamental yet diverse reasoning taxonomy underlying real-world frontier science, enabling more principled and effective scientific task construction.

References

- [1] Anthropic. Introducing claude sonnet 4.5, September 2025. URL <https://www.anthropic.com/news/claude-sonnet-4-5>.
- [2] Vladimir Baulin, Austin Cook, Daniel Friedman, Janna Lumiruuu, Andrew Pashea, Shagor Rahman, and Benedikt Waldeck. The discovery engine: A framework for ai-driven synthesis and navigation of scientific knowledge landscapes, 2025. URL <https://arxiv.org/abs/2505.17500>.
- [3] Jingyi Chai, Shuo Tang, Rui Ye, Yuwen Du, Xinyu Zhu, Mengcheng Zhou, Yanfeng Wang, Weinan E, Yuzhi Zhang, Linfeng Zhang, and Siheng Chen. Scimaster: Towards general-purpose scientific ai agents, part i. x-master as foundation: Can we lead on humanity’s last exam?, 2025. URL <https://arxiv.org/abs/2507.05241>.
- [4] Dave Citron. Try deep research and our new experimental model in gemini, your ai assistant, December 2024. URL <https://blog.google/products/gemini/google-gemini-deep-research/>.
- [5] Ryan Cory-Wright, Cristina Cornelio, Sanjeeb Dash, Bachir El Khadir, and Lior Horesh. Evolving scientific discovery by unifying data and background knowledge with ai hilbert. *Nature Communications*, 15(1):5922, July 2024. doi: 10.1038/s41467-024-50074-w.
- [6] Mingxuan Du, Benfeng Xu, Chiwei Zhu, Xiaorui Wang, and Zhendong Mao. Deepresearch bench: A comprehensive benchmark for deep research agents, 2025. URL <https://arxiv.org/abs/2506.11763>.
- [7] Kathryn Dumschott, Hannah Dörpholz, Marie-Angélique Laporte, Dominik Brillhaus, Andrea Schrader, Björn Usadel, Steffen Neumann, Elizabeth Arnaud, and Angela Kranz. Ontologies for increasing the fairness of plant research data. *Frontiers in Plant Science*, Volume 14 - 2023, 2023. ISSN 1664-462X. doi: 10.3389/fpls.2023.1279694. URL <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2023.1279694>.
- [8] Tianqing Fang, Hongming Zhang, Zhisong Zhang, Kaixin Ma, Wenhao Yu, Haitao Mi, and Dong Yu. Webevolver: Enhancing web agent self-improvement with coevolving world model, 2025. URL <https://arxiv.org/abs/2504.21024>.
- [9] Tianqing Fang, Zhisong Zhang, Xiaoyang Wang, Rui Wang, Can Qin, Yuxuan Wan, Jun-Yu Ma, Ce Zhang, Jiaqi Chen, Xiyun Li, Yonglin Wang, Jingchen Ni, Tianshi Zheng, Chun Chen, Wenhao Yu, Zhenwen Liang, Hongming Zhang, Haitao Mi, and Dong Yu. Cognitive kernel-pro: A framework for deep research agents and agent foundation models training, 2026. URL <https://arxiv.org/abs/2508.00414>.
- [10] FutureSearch, :, Nikos I. Bosse, Jon Evans, Robert G. Gambee, Daniel Hnyk, Peter Mühlbacher, Lawrence Phillips, Dan Schwarz, and Jack Wildman. Deep research bench: Evaluating ai web research agents, 2025. URL <https://arxiv.org/abs/2506.06287>.
- [11] Juraj Gottweis, Wei-Hung Weng, Alexander Daryin, Tao Tu, Anil Palepu, Petar Sirkovic, Artiom Myaskovsky, Felix Weissenberger, Keran Rong, Ryutaro Tanno, Khaled Saab, Dan Popovici, Jacob Blum, Fan Zhang, Katherine Chou, Avinatan Hassidim, Burak Gokturk, Amin Vahdat, Pushmeet Kohli, Yossi Matias, Andrew Carroll, Kavita Kulkarni, Nenad Tomasev, Yuan Guan, Vikram Dhillon, Eeshit Dhaval Vaishnav, Byron Lee, Tiago R D Costa, José R Penadés, Gary Peltz, Yunhan Xu, Ananisa Pawlosky, Alan Karthikesalingam, and Vivek Natarajan. Towards an ai co-scientist, 2025. URL <https://arxiv.org/abs/2502.18864>.
- [12] Minda Hu, Tianqing Fang, Jianshu Zhang, Junyu Ma, Zhisong Zhang, Jingyan Zhou, Hongming Zhang, Haitao Mi, Dong Yu, and Irwin King. Webcot: Enhancing web agent reasoning by reconstructing chain-of-thought in reflection, branching, and rollback, 2025. URL <https://arxiv.org/abs/2505.20013>.
- [13] Kexin Huang, Serena Zhang, Hanchen Wang, Yuanhao Qu, Yingzhou Lu, Yusuf Roohani, Ryan Li, Lin Qiu, Gavin Li, Junze Zhang, et al. Biomni: A general-purpose biomedical ai agent. *biorxiv*, 2025.
- [14] Zhengyao Jiang, Dominik Schmidt, Dhruv Srikanth, Dixing Xu, Ian Kaplan, Deniss Jacenko, and Yuxiang Wu. Aide: Ai-driven exploration in the space of code, 2025. URL <https://arxiv.org/abs/2502.13138>.
- [15] Andrej Karpathy. autoresearch: Ai agents running research on single-gpu nanochat training automatically. <https://github.com/karpathy/autoresearch>, 2026.
- [16] Kuan Li, Zhongwang Zhang, Huifeng Yin, Rui Ye, Yida Zhao, Liwen Zhang, Litu Ou, Dingchu Zhang, Xixi Wu, Jialong Wu, Xinyu Wang, Zile Qiao, Zhen Zhang, Yong Jiang, Pengjun Xie, Fei Huang, and Jingren Zhou. Websailor-v2: Bridging the chasm to proprietary agents via synthetic data and scalable reinforcement learning, 2025. URL <https://arxiv.org/abs/2509.13305>.

- [17] Kuan Li, Zhongwang Zhang, Huifeng Yin, Liwen Zhang, Litu Ou, Jialong Wu, Wenbiao Yin, Baixuan Li, Zhengwei Tao, Xinyu Wang, Weizhou Shen, Junkai Zhang, Dingchu Zhang, Xixi Wu, Yong Jiang, Ming Yan, Pengjun Xie, Fei Huang, and Jingren Zhou. Websailor: Navigating super-human reasoning for web agent, 2025. URL <https://arxiv.org/abs/2507.02592>.
- [18] Mukai Li, Qingcheng Zeng, Tianqing Fang, Zhenwen Liang, Linfeng Song, Qi Liu, Haitao Mi, and Dong Yu. Verified critical step optimization for llm agents, 2026. URL <https://arxiv.org/abs/2602.03412>.
- [19] Junteng Liu, Yunji Li, Chi Zhang, Jingyang Li, Aili Chen, Ke Ji, Weiyu Cheng, Zijia Wu, Chengyu Du, Qidi Xu, Jiayuan Song, Zhengmao Zhu, Wenhui Chen, Pengyu Zhao, and Junxian He. Webexplorer: Explore and evolve for training long-horizon web agents, 2025. URL <https://arxiv.org/abs/2509.06501>.
- [20] Zexi Liu, Yuzhu Cai, Xinyu Zhu, Yujie Zheng, Runkun Chen, Ying Wen, Yanfeng Wang, Weinan E, and Siheng Chen. Ml-master: Towards ai-for-ai via integration of exploration and reasoning, 2025. URL <https://arxiv.org/abs/2506.16499>.
- [21] Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The ai scientist: Towards fully automated open-ended scientific discovery, 2024. URL <https://arxiv.org/abs/2408.06292>.
- [22] Ziming Luo, Zonglin Yang, Zexin Xu, Wei Yang, and Xinya Du. Llm4sr: A survey on large language models for scientific research, 2025. URL <https://arxiv.org/abs/2501.04306>.
- [23] M-A-P, Xinrun Du, Yifan Yao, Kaijing Ma, Bingli Wang, Tianyu Zheng, King Zhu, Minghao Liu, Yiming Liang, Xiaolong Jin, Zhenlin Wei, Chujie Zheng, Kaixin Deng, Shawn Gavin, Shian Jia, Sichao Jiang, Yiyao Liao, Rui Li, Qinru Li, Sirun Li, Yizhi Li, Yunwen Li, David Ma, Yuansheng Ni, Haoran Que, Qiyao Wang, Zhoufutu Wen, Siwei Wu, Tyshawh Hsing, Ming Xu, Zhenzhu Yang, Zekun Moore Wang, Junteng Zhou, Yuelin Bai, Xingyuan Bu, Chenglin Cai, Liang Chen, Yifan Chen, Chengtuo Cheng, Tianhao Cheng, Keyi Ding, Siming Huang, Yun Huang, Yaoru Li, Yizhe Li, Zhaoqun Li, Tianhao Liang, Chengdong Lin, Hongquan Lin, Yinghao Ma, Tianyang Pang, Zhongyuan Peng, Zifan Peng, Qige Qi, Shi Qiu, Xingwei Qu, Shangaoran Quan, Yizhou Tan, Zili Wang, Chenqing Wang, Hao Wang, Yiya Wang, Yubo Wang, Jiajun Xu, Kexin Yang, Ruibin Yuan, Yuanhao Yue, Tianyang Zhan, Chun Zhang, Jinyang Zhang, Xiyue Zhang, Xingjian Zhang, Yue Zhang, Yongchi Zhao, Xiangyu Zheng, Chenghua Zhong, Yang Gao, Zhoujun Li, Dayiheng Liu, Qian Liu, Tianyu Liu, Shiwen Ni, Junran Peng, Yujia Qin, Wenbo Su, Guoyin Wang, Shi Wang, Jian Yang, Min Yang, Meng Cao, Xiang Yue, Zhaoxiang Zhang, Wangchunshu Zhou, Jiaheng Liu, Qunshu Lin, Wenhao Huang, and Ge Zhang. Supergpqa: Scaling llm evaluation across 285 graduate disciplines, 2025. URL <https://arxiv.org/abs/2502.14739>.
- [24] Grégoire Mialon, Clémentine Fourrier, Thomas Wolf, Yann LeCun, and Thomas Scialom. GAIA: a benchmark for general AI assistants. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024. URL <https://openreview.net/forum?id=fibxvavhs3>.
- [25] Ludovico Mitchener, Angela Yiu, Benjamin Chang, Mathieu Bourdenx, Tyler Nadolski, Arvis Sulovari, Eric C. Landsness, Daniel L. Barabasi, Siddharth Narayanan, Nicky Evans, Shriya Reddy, Martha Foiani, Aizad Kamal, Leah P. Shriver, Fang Cao, Asmamaw T. Wassie, Jon M. Laurent, Edwin Melville-Green, Mayk Caldas, Albert Bou, Kaleigh F. Roberts, Sladjana Zagorac, Timothy C. Orr, Miranda E. Orr, Kevin J. Zvezdaryk, Ali E. Ghareeb, Laurie McCoy, Bruna Gomes, Euan A. Ashley, Karen E. Duff, Tonio Buonassisi, Tom Rainforth, Randall J. Bateman, Michael Skarlinski, Samuel G. Rodrigues, Michaela M. Hinks, and Andrew D. White. Kosmos: An ai scientist for autonomous discovery, 2025. URL <https://arxiv.org/abs/2511.02824>.
- [26] Nayantara Mudur, Hao Cui, Subhashini Venugopalan, Paul Raccuglia, Michael P. Brenner, and Peter Norgaard. Feabench: Evaluating language models on multiphysics reasoning ability, 2025. URL <https://arxiv.org/abs/2504.06260>.
- [27] OpenAI. Introducing deep research. <https://openai.com/index/introducing-deep-research/>, February 2025.
- [28] Perplexity AI. Introducing perplexity deep research. <https://www.perplexity.ai/hub/blog/introducing-perplexity-deep-research>, 2025.
- [29] Long Phan, Alice Gatti, Nathaniel Li, Adam Khoja, Ryan Kim, Richard Ren, and Jason Hausenloy, et al. A benchmark of expert-level academic questions to assess ai capabilities. *Nature*, 649(8099): 1139–1146, January 2026. ISSN 1476-4687. doi: 10.1038/s41586-025-09962-4. URL <http://dx.doi.org/10.1038/s41586-025-09962-4>.

- [30] Samuel Schmidgall, Yusheng Su, Ze Wang, Ximeng Sun, Jialian Wu, Xiaodong Yu, Jiang Liu, Michael Moor, Zicheng Liu, and Emad Barsoum. Agent laboratory: Using llm agents as research assistants, 2025. URL <https://arxiv.org/abs/2501.04227>.
- [31] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024. URL <https://arxiv.org/abs/2402.03300>.
- [32] Zhihong Shen, Hao Ma, and Kuansan Wang. A web-scale system for scientific knowledge exploration. In Fei Liu and Thamar Solorio, editors, *Proceedings of ACL 2018, System Demonstrations*, pages 87–92, Melbourne, Australia, July 2018. Association for Computational Linguistics. doi: 10.18653/v1/P18-4015. URL <https://aclanthology.org/P18-4015/>.
- [33] Chenglei Si, Diyi Yang, and Tatsunori Hashimoto. Can llms generate novel research ideas? a large-scale human study with 100+ nlp researchers, 2024. URL <https://arxiv.org/abs/2409.04109>.
- [34] Xiangru Tang, Wanghan Xu, Yujie Wang, Zijie Guo, Daniel Shao, Jiapeng Chen, Cixuan Zhang, Ziyi Wang, Lixin Zhang, Guancheng Wan, Wenlong Zhang, Lei Bai, Zhenfei Yin, Philip Torr, Hanrui Wang, and Di Jin. Eigen-1: Adaptive multi-agent refinement with monitor-based rag for scientific reasoning, 2025. URL <https://arxiv.org/abs/2509.21193>.
- [35] Zhengwei Tao, Jialong Wu, Wenbiao Yin, Junkai Zhang, Baixuan Li, Haiyang Shen, Kuan Li, Liwen Zhang, Xinyu Wang, Yong Jiang, Pengjun Xie, Fei Huang, and Jingren Zhou. Webshaper: Agentically data synthesizing via information-seeking formalization, 2025. URL <https://arxiv.org/abs/2507.15061>.
- [36] MiroMind Team, Song Bai, Lidong Bing, Carson Chen, Guanzheng Chen, Yuntao Chen, Zhe Chen, Ziyi Chen, Jifeng Dai, Xuan Dong, Wenhan Dou, Yue Deng, Yunjie Fu, Junqi Ge, Chenxia Han, Tammy Huang, Zhenhang Huang, Jerry Jiao, Shilei Jiang, Tianyu Jiao, Xiaoqi Jian, Lei Lei, Ruilin Li, Ryan Luo, Tiantong Li, Xiang Lin, Ziyuan Liu, Zhiqi Li, Jie Ni, Qiang Ren, Pax Sun, Shiqian Su, Chenxin Tao, Bin Wang, Hellen Wang, Haonan Wang, James Wang, Jin Wang, Jojo Wang, Letian Wang, Shizun Wang, Weizhi Wang, Zixuan Wang, Jinfan Xu, Sen Xing, Chenyu Yang, Hai Ye, Jiaheng Yu, Yue Yu, Muyan Zhong, Tianchen Zhao, Xizhou Zhu, Yanpeng Zhou, Yifan Zhang, and Zhi Zhu. Mirothinker: Pushing the performance boundaries of open-source research agents via model, context, and interactive scaling, 2025. URL <https://arxiv.org/abs/2511.11793>.
- [37] Yuxuan Wan, Tianqing Fang, Zaitang Li, Yintong Huo, Wenxuan Wang, Haitao Mi, Dong Yu, and Michael R. Lyu. Inference-time scaling of verification: Self-evolving deep research agents via test-time rubric-guided verification, 2026. URL <https://arxiv.org/abs/2601.15808>.
- [38] Miles Wang, Robi Lin, Kat Hu, Joy Jiao, Neil Chowdhury, Ethan Chang, and Tejal Patwardhan. Frontierscience: Evaluating ai’s ability to perform expert-level scientific tasks, 2026. URL <https://arxiv.org/abs/2601.21165>.
- [39] Rui Wang, Ce Zhang, Jun-Yu Ma, Jianshu Zhang, Hongru Wang, Yi Chen, Boyang Xue, Tianqing Fang, Zhisong Zhang, Hongming Zhang, Haitao Mi, Dong Yu, and Kam-Fai Wong. Webaggregator: Enhancing compositional reasoning capabilities of deep research agent foundation models, 2026. URL <https://arxiv.org/abs/2510.14438>.
- [40] Xiaoxuan Wang, Ziniu Hu, Pan Lu, Yanqiao Zhu, Jieyu Zhang, Satyen Subramaniam, Arjun R. Loomba, Shichang Zhang, Yizhou Sun, and Wei Wang. Scibench: Evaluating college-level scientific problem-solving abilities of large language models, 2024. URL <https://arxiv.org/abs/2307.10635>.
- [41] Jason Wei, Zhiqing Sun, Spencer Papay, Scott McKinney, Jeffrey Han, Isa Fulford, Hyung Won Chung, Alex Tachard Passos, William Fedus, and Amelia Glaese. Browsecomp: A simple yet challenging benchmark for browsing agents, 2025. URL <https://arxiv.org/abs/2504.12516>.
- [42] Yixuan Weng, Minjun Zhu, Qiujie Xie, Qiyao Sun, Zhen Lin, Sifan Liu, and Yue Zhang. Deepscientist: Advancing frontier-pushing scientific findings progressively, 2025. URL <https://arxiv.org/abs/2509.26603>.
- [43] Andrew White, Michael Skarlinski, Jon Laurent, and Albert Bou. About 30% of humanity’s last exam chemistry/biology answers are likely wrong, July 2025. URL <https://www.futurehouse.org/research-announcements/hle-exam>.
- [44] Jialong Wu, Baixuan Li, Runnan Fang, Wenbiao Yin, Liwen Zhang, Zhengwei Tao, Dingchu Zhang, Zekun Xi, Gang Fu, Yong Jiang, Pengjun Xie, Fei Huang, and Jingren Zhou. Webdancer: Towards autonomous information seeking agency, 2025. URL <https://arxiv.org/abs/2505.22648>.

- [45] Jialong Wu, Wenbiao Yin, Yong Jiang, Zhenglin Wang, Zekun Xi, Runnan Fang, Linhai Zhang, Yulan He, Deyu Zhou, Pengjun Xie, and Fei Huang. Webwalker: Benchmarking llms in web traversal, 2025. URL <https://arxiv.org/abs/2501.07572>.
- [46] Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, et al. Autogen: Enabling next-gen llm applications via multi-agent conversations. In *First conference on language modeling*, 2024.
- [47] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report, 2025. URL <https://arxiv.org/abs/2505.09388>.
- [48] Zhongyue Zhang, Zijie Qiu, Yingcheng Wu, Shuya Li, Dingyan Wang, Zhuomin Zhou, Duo An, Yuhao Chen, Yu Li, Yongbo Wang, Chubin Ou, Zichen Wang, Jack Xiaoyu Chen, Bo Zhang, Yusong Hu, Wenxin Zhang, Zhijian Wei, Runze Ma, Qingwu Liu, Bo Dong, Yuexi He, Qiantai Feng, Lei Bai, Qiang Gao, Siqi Sun, and Shuangjia Zheng. Origene: A self-evolving virtual disease biologist automating therapeutic target discovery. *bioRxiv*, 2025. doi: 10.1101/2025.06.03.657658. URL <https://www.biorxiv.org/content/early/2025/06/06/2025.06.03.657658>.
- [49] Zixuan Zhang, Nikolaus Parulian, Heng Ji, Ahmed Elsayed, Skatje Myers, and Martha Palmer. Fine-grained information extraction from biomedical literature based on knowledge-enriched Abstract Meaning Representation. In Chengqing Zong, Fei Xia, Wenjie Li, and Roberto Navigli, editors, *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 6261–6270, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.acl-long.489. URL <https://aclanthology.org/2021.acl-long.489/>.
- [50] Tianshi Zheng, Zheyang Deng, Hong Ting Tsang, Weiqi Wang, Jiaxin Bai, Zihao Wang, and Yangqiu Song. From automation to autonomy: A survey on large language models in scientific discovery. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 17733–17750, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-332-6. doi: 10.18653/v1/2025.emnlp-main.895. URL <https://aclanthology.org/2025.emnlp-main.895/>.
- [51] Tianshi Zheng, Kelvin Kiu-Wai Tam, Newt Hue-Nam K. Nguyen, Baixuan Xu, Zhaowei Wang, Jiayang Cheng, Hong Ting Tsang, Weiqi Wang, Jiaxin Bai, Tianqing Fang, Yangqiu Song, Ginny Y. Wong, and Simon See. Newtonbench: Benchmarking generalizable scientific law discovery in llm agents, 2026. URL <https://arxiv.org/abs/2510.07172>.

A Technical Details

Agent Framework Details Our agent framework is built upon Cognitive Kernel-Pro [9], a two-tier, multi-module hierarchical architecture in which a main agent is responsible for task decomposition, subtask delegation, evidence aggregation, tool invocation, and Python-based action generation, while specialized sub-agents handle grounded interactions with external environments. Concretely, we instantiate a web agent for live web navigation and a file agent for local document processing, following the general design of Cognitive Kernel-Pro. Different from the original framework, however, we remove the `ask_llm` function from the main agent. This modification prevents the agent from bypassing tool use and producing unsupported shortcut answers directly from the base language model, thereby encouraging explicit evidence collection through interaction with the web and files. To further adapt our agent for frontier scientific reasoning, we refine the system prompt of the main agent to encourage proactive acquisition of domain knowledge and careful verification.

Agent	Action / Tool	Observation
Main	<code>web_agent(task)</code>	Web-agent output / log
	<code>file_agent(task)</code>	File-agent output / log
	<code>simple_web_search(query)</code>	Search results
	<code>stop(answer, summary)</code>	Final answer
Web	<code>click(id)</code>	Web text & DOM
	<code>type(id, text)</code>	Web text & DOM
	<code>scroll_up()</code> / <code>scroll_down()</code>	Updated page state
	<code>wait()</code>	Updated page state
	<code>goback()</code>	Previous page state
	<code>restart()</code>	Reset browser state
	<code>goto(url)</code>	Web text & DOM
	<code>save(remote, local)</code>	Saved local file / status
	<code>screenshot(flag, path)</code>	Screenshot-enabled page view
	<code>stop(answer, summary)</code>	Final sub-agent output
File	<code>load_file(path)</code>	File metadata / page index
	<code>read_text(path, pages)</code>	Extracted file text
	<code>read_screenshot(path, pages)</code>	Visual page content
	<code>search(path, keywords)</code>	Matched spans / page hits
	<code>stop(answer, summary)</code>	Final sub-agent output

Table 5: Tool and action space of our agent framework.

Agent Backbone Configuration Across all experiments, we optimize only the main agent and use Qwen3-8B (without thinking) as its backbone. The sub-agents—specifically, the web agent and the file agent—remain frozen and use Qwen3-32B (without thinking) throughout trajectory sampling, reinforcement learning, and benchmark evaluation.

B Further Details of SciResearcherQA

In this section, we include consolidated prompt templates for the four core modules in the construction of SciResearcherQA.

Prompt template for Conceptual Task - seed2question.

Conceptual Task - seed2question

```
## Core Identity
<Role: Frontier Scientific Question Curation Agent. Your goal is to generate a
base conceptual scientific reasoning question from a given seed entity, grounded
in verifiable academic evidence. The target domains include frontier scientific
areas such as biology, chemistry, biomedicine, and related interdisciplinary
fields.>

## Input Format
<Seed entity, together with its domain and ontology information; illustrative
examples.>

## Question Curation Requirements
<Metric Definition>
<The generated question should:
1. Include the seed entity or be directly grounded in it.
2. Be concise but scientifically meaningful.
3. Be answerable from a single authoritative academic source at this stage.
4. Prefer multiple-choice format with plausible confounders, while allowing
short-answer format when more appropriate.
5. Avoid shortcuts that can be solved by trivia, superficial keyword matching,
or generic web search without reading the academic evidence.
6. Be suitable as the semantic backbone for later anchor-based augmentation.>

## Pre-Action Protocol: Plan Before Searching
<Metric Definition>
<Before browsing, understand the seed entity and its scientific context. Plan
3--5 diverse search queries that target academic sources such as peer-reviewed
papers, domain databases, preprints, and reputable scientific venues. Assess
source quality based on relevance, authority, evidence specificity, and whether
the source supports a nontrivial scientific claim.>

## Question Curation Strategies
<Metric Definition>
<Scoring Description and Examples for each numbered strategy:>
1. Meticulousness and persistence in finding high-quality academic evidence.
2. Task decomposition: search -> evidence extraction -> question generation ->
verification.
3. Adaptive error handling and reuse of progress state when searches fail or
evidence is insufficient.
4. Multi-query scout search and URL selection based on relevance, venue quality,
source diversity, and scientific specificity.
5. Use of the url2evidence sub-agent to access selected academic sources,
extract key supporting evidence, and distinguish stand-alone scientific facts
from study-specific artifacts.
6. Evidence quality checks, including source authority, evidence-answer
entailment, and avoidance of unsupported assumptions.
7. Question formulation with plausible, unbiased, and challenging confounders
for MCQs; clear expected answer for short-answer questions; and final
quality checks.
8. Multi-tool coordination following the typical workflow:
scout search -> source selection -> url2evidence -> question generation ->
verification.

## Output Format
The final output MUST be a JSON object with the following structure:
'''json
{
  "question": "The question text containing or directly grounded in the seed entity",
```

```

"answer": "The correct answer content, not a letter label",
"question_type": "mcq",
"confounders": ["confounder1", "confounder2", "confounder3"],
"evidence": {
  "url": "https://...",
  "paper_title": "Title of the paper or academic source",
  "evidence_paragraph": "The exact paragraph or quote that supports the answer",
  "context": "Additional background information that explains the broader context of
the evidence or clarifies details needed to understand it"
}
}
'''

```

****Important Notes on Output Format:****

- 'answer': Provide the actual answer content, not an option letter.
- 'question_type': Use lowercase "mcq" or "short_answer".
- 'confounders': For MCQs, provide 3 or more challenging wrong answers.
- 'evidence.url': Must be a real, verified URL.
- The entire output must be valid JSON and contain no extraneous commentary.

Prompt template for Conceptual Task - anchor_extraction.

Conceptual Task - anchor_extraction

You are an expert at analyzing scientific reasoning questions.

Your task is to identify the single most critical "anchor entity" in the question body. This anchor will be used for anchor-based question augmentation: a new question will later be generated whose answer is exactly this anchor entity, and that new question will be fused back into the original question.

Definition of Anchor Entity

An anchor entity is a SPECIFIC scientific term that:

1. ****Domain-specific****: It is a concrete scientific entity, such as a gene, protein, pathway, compound, species, technique, disease, mutation, phenotype, material, model, or other scientific concept.
2. ****Question-body only****: It appears in the question stem but does NOT appear in the correct answer or any confounder.
3. ****Decisive****: The question becomes substantially harder or unanswerable if this entity is masked or removed.
4. ****Specific and concrete****: It is sufficiently specific to support further evidence-grounded browsing and question generation.

Your Task

Given the question, correct answer(s), and confounders below, you must:

1. Identify candidate anchor entities in the question body.
2. Verify that each candidate does NOT appear in the correct answer or any confounder.
3. Evaluate whether each candidate is decisive for deriving the final answer.
4. Select the most decisive, specific, and concrete entity.
5. If no valid anchor exists, return an empty string.

Selection Criteria (in priority order)

1. Prefer the MOST SPECIFIC entity, e.g., "AXL" over "receptor tyrosine kinase".
2. Prefer entities that constrain the answer, such that removing them makes multiple answers plausible.
3. Prefer named entities, such as gene, protein, compound, disease, pathway, or model names, over generic scientific terms.
4. Prefer entities that are decoupled from the surface form of the answer options.
5. If multiple candidates exist, choose the one most central to the scientific claim.

Output Format

Return ONLY valid JSON:

```
{
  "candidates": [
    {
      "entity": "...",
      "in_question": true,
      "in_options": false,
      "is_decisive": true
    }
  ],
  "anchor_entity": "<the single valid anchor entity string, or empty string if none>",
  "entity_type": "<type: gene|protein|pathway|compound|technique|disease|other>",
  "reasoning": "<brief explanation of selection and validation>"
}
```

Examples

<Illustrative worked examples omitted here for brevity.>

```
## Question
{question}

## Correct Answer(s)
{answer}

## Confounders (wrong options)
{confounders}

Analyze candidates, verify constraints, and return the valid anchor entity.
```

Prompt template for Computational Task - seed2equation.

Computational Task - seed2equation

```
## Core Identity
<Frontier Scientific Model Discovery Agent. Your goal is to identify an advanced
computational or numerical scientific model associated with a given seed entity,
extract its governing equations and application constraints from academic
sources, and prepare the model specification for downstream computational
question generation.>

## CARDINAL RULE: Precision and Groundedness Above All
<Metric Definition>
<All extracted model details must be traceable to the selected academic source.
Prefer precise, reproducible mathematical definitions over vague model
descriptions. Abandon a candidate model if its equations, parameters, scenario,
or source cannot be verified.>

## Input Format
<Seed entity, together with its domain and ontology information; illustrative
examples.>

## Task Overview -- Three-Level Evidence Selection and Model Extraction

### Level 1: Scout Search
Perform multiple scout searches to identify promising academic sources associated
with the seed entity. Use diverse queries that target computational models,
numerical simulations, mechanistic equations, kinetic models, ODE/PDE systems,
statistical models, or other quantitative formulations.

### Level 2: URL Evaluation with eval_urls
Use the eval_urls tool to assess selected sources. Prioritize sources according
to:
1. Model exclusiveness
2. Search identifiability
3. Computational complexity
4. LLM unfamiliarity

Also consider URL validity and whether the source clearly contains a usable
computational or numerical model.

### Level 3: Detailed Model Extraction with url2evidence
Use the url2evidence sub-agent to conduct a deep dive into the final selected
source or sources. Extract the complete model specification, including:
1. Model name and scientific purpose.
2. Governing equations.
3. Variable definitions.
4. Parameter definitions and units.
5. Applicable scenario and constraints.
6. Any assumptions required for correct model use.

## Model Selection Criteria

Select a model that satisfies as many of the following criteria as possible:
1. The model supports calculable numerical outputs.
2. The model is described in a real, citable academic source.
3. The equations are nontrivial and not merely standard textbook formulas.
4. The computation requires meaningful model instantiation or numerical solving.
5. The model can support a realistic scenario-based scientific question.
6. The source is relatively recent, niche, or unlikely to be memorized by LLMs.
7. The model is clearly associated with the seed entity or its scientific domain.

## What Counts as a Frontier Numerical Model?
<A model with explicit mathematical structure, such as governing equations,
ODE/PDE systems, kinetic models, dose-response models, mechanistic simulations,
quantitative biological or chemical models, or other computational formulations
that can be instantiated to produce a numerical answer.>
```

```

## What Does NOT Count
<Do not select models that are only conceptual diagrams, purely descriptive
frameworks, simple textbook equations, standard unit conversions, or models whose
parameters and equations cannot be verified from the source.>

## Output Format
The final output MUST be a JSON object with the following structure:
'''json
{
  "seed_entity": "<the seed entity>",
  "selected_model": {
    "title": "<paper title>",
    "url": "<URL of the paper or academic source where the model is described>",
    "description": "<brieﬀ description and its applicable scenario>",
    "equations": "<explicit named equations; see Equation Naming Convention below>",
    "variables": "<deﬂnitions of variables used in the equations>",
    "parameters": "<deﬂnitions, values if available, and units of parameters>",
    "assumptions": "<model assumptions and application constraints>"
  }
}
'''

**Critical Notes on Output:**
- 'selected_model.url' MUST be a real, veriﬂed URL, not a fabricated one.
- 'equations' MUST contain explicit mathematical deﬂnitions, not vague
descriptions.
- Include all variables, parameters, units, assumptions, and constraints needed
for downstream question generation.
- The entire output must be parsable as JSON and contain no extraneous text or
commentary.

## Equation Naming Convention
<Use bracketed 5--20 word names for each equation. Each name should describe the
scientiﬂc role of the equation, e.g., [Logistic tumor growth with drug-induced
cell damage]. Avoid generic names such as [Equation 1] or [Main formula].>

```

Prompt template for Computational Task - eval_urls.

Computational Task - eval_urls

You are an expert scientific model evaluator. You are given the text content of a scientific article or paper. Your task is to evaluate whether this article contains a computational or numerical model suitable for generating a benchmark question that tests an AI agent's ability to:

1. Search for and identify the relevant model.
2. Extract the model equations and constraints from the paper.
3. Instantiate the model in a concrete scientific scenario.
4. Write and execute a Python solver to compute a numerical answer.

First, perform preliminary validity checks. Then evaluate the article according to the four core metrics used for computational task curation.

Preliminary Check 1: URL Validity

<Metric Definition>

<Determine whether the URL corresponds to a real and accessible academic source, such as a peer-reviewed paper, preprint, official proceedings page, or reputable scientific database entry.>

Preliminary Check 2: Model Presence

<Metric Definition>

<Determine whether the article contains an explicit computational or numerical model with equations, variables, parameters, or algorithmic procedures that can be used to compute a numerical answer.>

Core Metric 1: Model Exclusiveness (0-10) -- CRITICAL

<Metric Definition>

<Score how specific and source-dependent the model is. High-scoring models have equations, assumptions, or parameterizations that are distinctive to this paper or research line, rather than generic textbook formulas.>

Core Metric 2: Search Identifiability (0-10)

<Metric Definition>

<Score whether an agent could plausibly find the source through web search from the seed entity, scientific context, or model-related clues. A good source should be searchable but not trivially obvious.>

Core Metric 3: Computational Complexity (0-10)

<Metric Definition>

<Score whether the model requires nontrivial quantitative reasoning, numerical simulation, equation solving, or careful parameter instantiation. Avoid models that require only simple arithmetic or direct lookup.>

Core Metric 4: LLM Unfamiliarity (0-10)

<Metric Definition>

<Score how unlikely the model and its exact equations are to be memorized by a general-purpose LLM. Niche, recent, specialized, or paper-specific models should receive higher scores.>

Output Format -- strict JSON

```
'''json
{
  "is_valid_url": true,
  "includes_model": true,
  "model_exclusiveness": 8,
  "search_identifiability": 7,
  "computational_complexity": 8,
  "llm_unfamiliarity": 9,
  "model_name": "<name of the model if identifiable, else 'N/A'>",
  "model_summary": "<1-2 sentence summary of what the model computes>",
  "rationale": "<brief rationale for your validity checks and scores>"
}
'''
```

Return ONLY the JSON object, with no commentary before or after.